Supplying Nitrogen on Future Missions to the Moon and Mars

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Resupply efforts of life support consumables will become increasingly more difficult as future missions are conducted further from Earth. Not only will travel take days, months or even years to reach a mission's location, mission durations are expected to increase, thereby significantly increasing the quantity of life support consumables necessary. This situation poses a major challenge for supplying nitrogen specifically because it is not a currently recycled resource. Nitrogen is primarily used to maintain cabin pressure in vehicles/habitats and is lost nominally to the environment through leaks, airlock usage, and technology venting. Therefore, while not being a metabolic consumable, nitrogen has been vital to maintaining a habitable environment. This work details a nitrogen supply trade study conducted across all currently known future missions. Nitrogen needs were evaluated across missions for Orion, Gateway, Human Landing System (HLS), Pressurized Rover (PR), lunar surface habitat, mars transit, and mars surface habitat. Both nominal and contingency cases were considered for each habitat/vehicle to include an upper limit on nitrogen needs. Potential nitrogen supply methods were sized according to the nitrogen needs of each evaluated mission and were compared against conventional compressed gas tanks using an equivalent system mass (ESM) analysis. The missions are presented here in aggregate and showcase the mass savings associated with non-gaseous options as nitrogen usage needs increase. Understanding the nitrogen usage needs of future vehicles/habitats and the potential methods to meet those needs is necessary to inform logistical planning, vehicle design and technology developments for future missions.

Acronyms and Nomenclature

BVAD	=	Baseline Values and Assumptions Document	LEO	=	Low Earth Orbit
CFC	=	Cryogenic flux capacitor	MLI	=	Multi-Layer Insulation
CH4	=	methane	MPH	=	Multi-Purpose Habitat
CO_2	=	carbon dioxide	NORS	=	Nitrogen/Oxygen Recharge System
ESM	=	Equivalent System Mass	NRHO	=	Near Rectilinear Halo Orbit
EVA	=	Extravehicular Activity	N_2	=	nitrogen
H_2O	=	water	N_2H_4	=	hydrazine
HLS	=	Human Landing System	N_2O	=	nitrous oxide
HPGC	=	High pressure gas carrier	O_2	=	oxygen
ISRU	=	In-Situ Resource Utilization	PR	=	Pressurized Rover
ISS	=	International Space Station	SCN2	=	Supercritical nitrogen
JAXA	=	Japanese Aerospace Exploration Agency			

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I. Introduction

As space missions progress toward longer duration missions that are further and further from Earth, resupply efforts will become increasingly more difficult. Not only will missions well beyond low earth orbit (LEO) take days, months or even years to reach, there is the potential for multiple vehicles/habitats to be in use at once [1]. Such multi-mission far-from-Earth scenarios present a major challenge, particularly for re-supply efforts and logistical planning of vital life support consumables. Life support consumables like oxygen (O_2), water (H_2O), and nitrogen (N_2), are consumed in large quantities and require large volumes to store, making them difficult to house for long periods of time. Efforts have been made to reduce resupply and storage of O_2 and H_2O by recycling and electrolyzing H_2O – a method currently employed on the International Space Station (ISS). However, N_2 is not a currently recycled resource and though it is not consumed metabolically, it is consistently lost to the environment.

N_2 has historically been used to maintain cabin pressure in vehicles and habitats, akin to creating an Earth-like atmosphere. When a portion of the vehicle/habitat atmosphere is lost to space, N_2 is lost. These losses can occur from nominal leakages, airlock usage, venting other life support technologies (e.g., CO_2/H_2O removal technologies), and/or contingency events (e.g., off-nominal leaks, repress due to fire, etc.). Airlock usage in particular is expected to play a major role in the N_2 usage of future surface missions where frequent extravehicular activity (EVA) results in large N_2 usage due to the frequency in which the airlock is depressed/ repressed. Additional advances in habitation could also result in further increases in N_2 usage through activities like plant growth and/or science.

Compressed gas cylinders have traditionally been used to supply N_2 because they are relatively easy to manage and have well-defined risks in space vehicles. However, compressed gas cylinders are neither efficient in mass nor volume as N_2 needs increase. Furthermore, there has been a limited understanding of what the N_2 needs of future missions will be since previous focus has remained with metabolic consumables. This report details a trade study conducted on N_2 needs across future missions and potential supply options using an equivalent system mass (ESM) analysis. The results of this trade inform technology developments necessary for future missions.

II. Study Approach

Relevant information on future vehicles/habitats, N_2 needs, and N_2 supply methods was gathered from a wide variety of sources, including public literature searches and NASA internal conversations. This information was filtered to finalize the bounds of the trade and perform the necessary calculations to determine N_2 usage needs and technology sizing, which is further detailed below.

A. Evaluation of N_2 Needs for Pressure Control

N_2 needs were evaluated for currently planned future missions, defined as those starting from ~Artemis 6 to long term habitation on Mars. The choice of Artemis 6 refers to when all Gateway modules are expected to be assembled around the Moon in Near Rectilinear Halo Orbit (NRHO) [2]. This definition of future missions was defined to focus efforts primarily on long-duration missions to the Moon and Mars where N_2 needs are expected to increase significantly and resupply will be costly. N_2 usage is defined for this work as those needs pertaining to pressure control. Since other possible needs related to advanced habitation and science activities that may exist in the future are not currently defined, these were not included. The following vehicle and habitats were assessed for N_2 usage across missions that fit within the future mission definition:

- Orion
- Gateway
- Human Landing System (HLS)
- Pressurized Rover (PR)
- Multi-Purpose Habitat (MPH)
- Mars transit vehicle
- Mars surface habitat

N_2 usage for pressure control depends on the concept of operations of the vehicle/habitat and mission but was generally attributed to the following: nominal leaks of the vehicle/habitat, N_2 lost to venting technologies (e.g., CO_2 and H_2O removal technologies), airlock usage (i.e., EVA frequency), and contingency events that lose cabin air (e.g., an unplanned leak, repress of cabin, fire, etc.). For this work, each vehicle/habitat was assessed for a minimum of two missions to bound the vehicle/habitat's upper and lower N_2 usage. The N_2 needs were either provided by the relevant program, or the appropriate information was gathered to estimate N_2 usage for the respective missions of the vehicle/habitat. In some cases, a parametric analysis was necessary to estimate the bounds of N_2 needs since several vehicle/habitat details are not yet defined, e.g. the volume of a Mars transit vehicle. For these situations, a reasonable range for each unknown parameter was provided by the program or pulled from the Life Support Baseline Values and Assumptions Document (BVAD) [3]. The N_2 usage needs acquired or calculated were fed into the remainder of the

trade – sizing technologies according to the respective N_2 needs. For every vehicle/habitat, at least one mission case containing contingency N_2 was included to better estimate the upper limit of N_2 needs. Contingency cases primarily included scenarios requiring a full cabin repress or feeding a leak.

B. Selection & Sizing of N_2 Supply Options

Research was conducted to identify N_2 supply technologies that have potential use on future missions. The pool was initially made broad to include both technologies that have flown and those still under development. This broader list included any method that could store, recycle, or harvest nitrogen in any form, including fixed nitrogen [4], [5], [6]. This list was later reduced to options that could supply diatomic nitrogen specifically for pressure control. Technologies with limited information that would be difficult to accurately size were also removed at that point, e.g. ionic liquids whose development is still in the fundamental stages [7]. The final options below were chosen based on available information and internal conversations with subject matter experts.

- Compressed gas
- Liquid N_2
- Supercritical N_2
- N_2O decomposition
- N_2H_4 decomposition
- Cryogenic flux capacitor
- Harvesting N_2 on Mars

The supply methods include several different compressed gas cylinders, methods of generating N_2 from liquids with a higher boiling point, a newer technology called a cryogenic flux capacitor (CFC) that utilizes physisorption to store N_2 in solid-like state [8], and In-Situ Resource Utilization (ISRU) of N_2 on Mars using preparative gas chromatography. Mass, power (if applicable), and volume were sized for each technology for every mission case considered within each vehicle/habitat considered. This mass, power, and volume were later fed into the equivalent system mass (ESM) analysis to estimate mass, power, volume, and cooling impacts on the vehicle/habitat. Sizing was dependent on each technology's concept of operations but typically included at a minimum, a pressure vessel to hold the fluid and parts to remove the fluid from the tank and supply it to the cabin as gaseous N_2. Each technology and its respective sizing details and assumptions are found in the following subsections.

1. *Compressed Gas*

Four different compressed gas cylinders were initially considered to account for the difference between those used previously and those being planned for future missions. These consisted of Nitrogen/Oxygen Recharge System (NORS) tanks - currently supply compressed gas on the ISS [9], ARDE tanks - smaller mass to capacity ratio tanks developed by ARDE, Inc. and are being planned for future missions, Japanese Aerospace Exploration Agency (JAXA) tanks - similar to ARDE and also being planned on future missions, and the high pressure gas carrier (HPGC) - a concept tank modeled after NORS for general exploration [10]. These options were compared against one another to determine a single baseline to compare all other N_2 supply options against.

Tank sizing was conducted by estimating the number of tanks (N_{tanks}) needed to meet the N_2 usage needs for each mission based on the useful capacity of the tank. Partially filled tanks were not considered. If the mission requirements exceeded one full tank of N_2 but would not completely fill a second tank, two full tanks were assumed necessary for the mission. The total mass per tank included the mass of the tank, mass of nitrogen within the tank (including ullage), mass of a regulator, mass of recharge valve assembly and mass of packaging necessary to launch the tank. Packaging was made to be consistent with tanks currently packaged for delivery to ISS where they are wrapped in foam and placed in an M01 bag [10]. Volume for each tank was assumed to be the external volume of the M01 bag. No power was considered for the compressed gas tank options. A schematic representing the compressed gas tank sizing and elucidating useful capacity is presented in Figure 1.

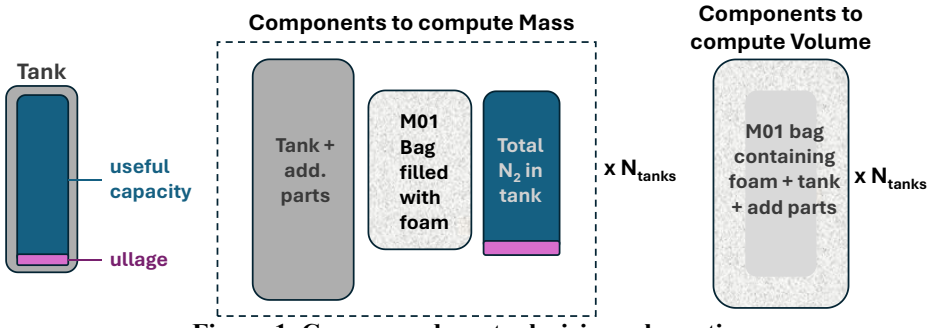


Figure 1. Compressed gas tank sizing schematic.

2. Cryogenic Fluids – Liquid N_2 & Supercritical N_2

Liquid and supercritical forms of N_2 have a higher density than compressed gas and therefore have the potential to reduce mass and volume on future long-term missions simply because more N_2 can be stored in a smaller volume. However, cryogenic systems are more complex than compressed gas tanks and require components additional to the tank to maintain the cryogenic fluid. For this work, in addition to a tank, components necessary for heat removal and vaporization of N_2 were considered based on the type of fluid and concept of heat removal. For supercritical N_2, heat was only assumed to be removed with the use of a cryocooler (power) and did not require additional components to vaporize since gaseous N_2 can be pulled directly from the tank. For liquid N_2, heat could be removed by boiling off some of the N_2 (boil-off) or again using power to cool via a cryocooler. Liquid N_2 was also assumed to require a cryopump and vaporizer to supply gaseous N_2. Figure 2 depicts a flow chart schematic of the components included in sizing liquid and supercritical N_2 options based on the type of fluid and method of heat removal. Components are not to scale and include the N_2 separate from the tank for completeness.

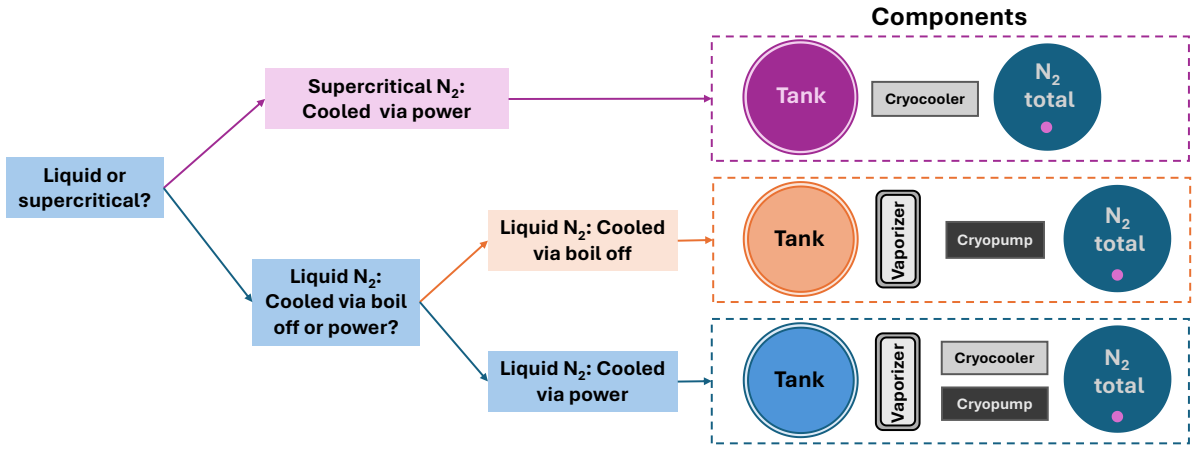


Figure 2. Flow chart schematic of components included in the sizing of liquid N_2 and supercritical N_2 supply options.

The cryocooler is based on the Sunpower Cryotel CT [11]. The vaporizer is a concept vaporizer composed of an ARDE tank attached to a drum heater and the cryopump is based on a Barber Nichols liquid O_2/CH_4 pump. The cryogenic tank was sized using a cryogenic tank sizing tool developed in MATLAB and includes the pressure vessel, penetrations, valves and support structure, multi-layer insulation (MLI), tank supports, and a vacuum jacket. The tank system sizing depends on the cryogenic fluid, mass of N_2 required for the mission, heat removal concept, mission duration, and environmental conditions. The tool, shown schematically in Figure 3, computes the total tank system mass and volume, total cryogenic fluid mass (accounting for ullage and boil-off, if applicable), and power required to maintain the fluid if the power option is chosen [12]. All inputs to the tool are provided in a Microsoft Excel worksheet, where missions are run sequentially by iterating on total cryogenic mass or power until the heat requirement for each scenario is met. For this work, all cryogenic tanks were assumed to be a spherical, composite tank with 50 layers of MLI between the pressure vessel and surrounding vacuum jacket. For each mission, a single tank was sized to accommodate the necessary cryogenic fluid needed to meet the N_2 needs of the mission. Liquid N_2 was assumed stored at its boiling point (-196°C) and an elevated pressure of 50 psia, while supercritical N_2 was assumed stored at -145°C and 1000 psia.

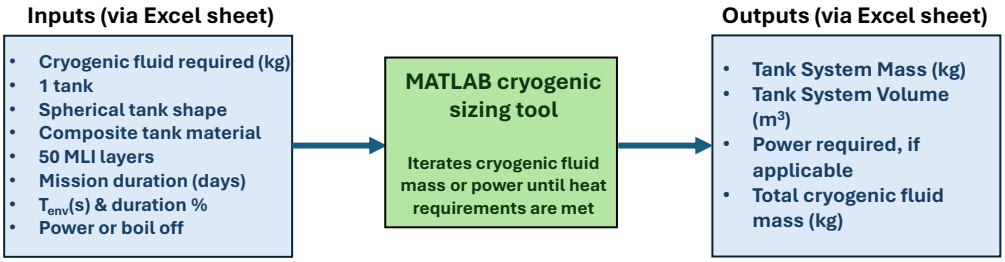


Figure 3. Cryogenic Tank Sizing Tool Schematic.

Relevant environmental temperatures (Table 1) necessary for accurate sizing using the cryogenic sizing tool were determined for each mission. Every mission assumes at least two temperatures, corresponding to a location with day and night (surface) or sun and shade (transit). In most cases, the temperatures considered were split evenly across the mission duration, corresponding to the assumption that cold temperatures are experienced equally as much as hot temperatures. HLS is the only exception to this rule because the amount of time the vehicle spends in NRHO versus the lunar surface depends on the type of mission (sortie vs excursion) and whether the vehicle is assumed reused or not. For Orion, Gateway, and the Mars Transit vehicle, their actual locations will vary as they move through space but their thermal environment was assumed equivalent to the values assumed for NRHO. PR was assumed to traverse across the lunar surface, experiencing evenly both nominal and extreme hot and cold temperatures expected for potential EVA locations. MPH was assumed to reside on the lunar plain where more nominal temperatures are expected. The Gale crater was assumed for the Mars surface habitat location, where temperature data exists for day/night and winter/summer. Gale crater is also near the Martian equator where temperatures are less extreme and may be more conducive to a long term surface stay.

Table 1. Environmental temperatures assumed for cryogenic tank sizing per vehicle/habitat.

Vehicle/ Habitat	Assumed Environment	Environmental Conditions (# Temperatures)
Orion	NRHO	Full sun/shade (2)
HLS	NRHO/Lunar Surface	Full sun/shade & lunar plain hot/cold (4)
Gateway	NRHO	Full sun/shade (2)
PR	Lunar Surface	Lunar plain hot/cold & Lunar extreme hot/cold (4)
MPH	Lunar Surface	Lunar plain hot/cold (2)
Mars Surface Habitat	Mars Surface	Gale Crater day/night and winter/summer (4) [20]
Mars Transit Vehicle	NRHO	Full sun/shade (2)

3. N_2 Generation from N_2O and N_2H_4

N_2 bound within nitrous oxide (N_2O) or hydrazine (N_2H_4) can be stored as a high-density liquid with a higher boiling point than cryogenic N_2 options and then chemically processed into low-density N_2 gas when needed. The benefit of this generation scheme is that the initial form (N_2O or N_2H_4) requires far less heat removal and this can have impacts on the overall system size, in addition to being easier to maintain. Additionally, since nitrous oxide can be catalytically broken down into N_2 and O_2, and hydrazine can be catalytically broken down into N_2 and H_2, these supply methods have the potential to supplement fuel or life support systems with the remaining reaction molecules. Of course, this comes at the cost of needing to bring additional equipment to chemically convert the original liquid into N_2 gas pure enough to release into the cabin. There can also be additional safety concerns should the original fluid spur a leak.

The general sizing considerations made for N_2 generation technologies are schematically represented in Figure 4. For both N_2O and N_2H_4, boil-off or power-based cooling methods are considered. To maintain N_2O in a liquid state, it is assumed to be held at its boiling point (-89°C) and an elevated pressure of 50 psia. Similarly, N_2H_4 is assumed held at its boiling point (114°C) and an elevated pressure of 50 psia. The tank and total N_2O / N_2H_4 are determined with the cryogenic tank sizing tool since both pressure and temperature must be maintained to ensure a liquid state. The same cryocooler listed for cryogenic N_2 is included for the power-based cooling method. The production unit

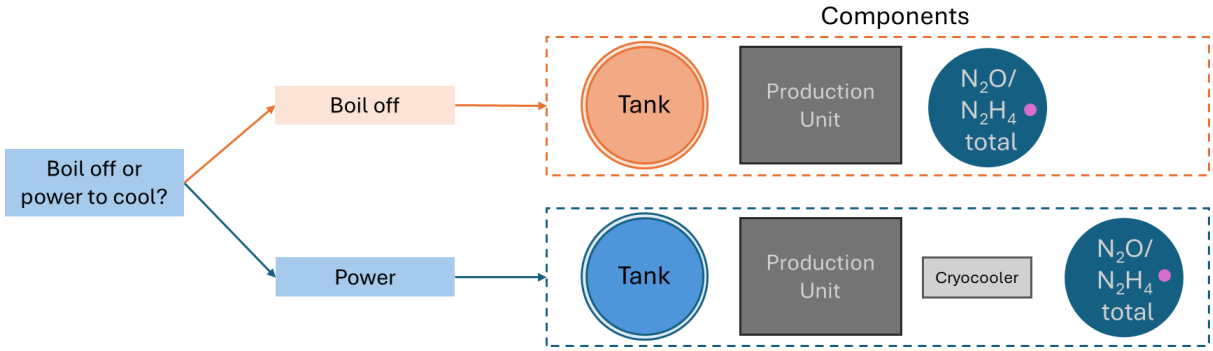


Figure 4. Flow chart schematic for N_2 generation technology sizing.

differs based on the fluid and is assumed to include all other necessary components to pull liquid out of the tank and process it into purified N_2. The N_2O production unit is based on a commercially available mobile unit for hospitals that decomposes N_2O into N_2 and O_2 via a catalytic process [13]. The N_2H_4 production unit is based on a system developed by Life Systems Inc. under a NASA contract [14]. For both units, it was assumed the N_2 production from N_2O had no losses. The remaining O_2/H_2 was assumed a mass benefit to each respective supply method because they could be used within the mission to supplement metabolic O_2, fuel, and/or aid a Sabatier in reducing metabolic CO_2 to CH_4. This mass benefit inherently assumes all systems necessary to process the O_2/H_2 are already present within the respective vehicle.

4. Cryogenic Flux Capacitor (CFC)

The cryogenic flux capacitor (CFC) is a technology under development that aims to find the middle ground between compressed gases and cryogenic fluids through physisorption. By holding the fluid molecules in a solid-state like condition within an adsorbent material, CFC's can store fluids at high densities while operating at relatively low pressures and relatively high temperatures, above the liquid's boiling point. The fluid can then be released upon a change in temperature or pressure. A typical CFC is composed of an adsorbent blanket that holds the fluid – setting the capacity of the device, layered with a thermally conductive membrane that is typically corrugated to allow for efficient fluid delivery [8] [15]. The layered device is placed inside a pressure boundary and wrapped in thermal insulation. To release the fluid, the thermally conductive membrane is conducted to a heat supply unit.

For this work, the adsorbent chosen is an aerogel based on Cryogel, [8] the thermally conductive membrane is corrugated aluminum, the pressure boundary and thermal insulation is a cryogenic tank, and the heat supply unit is a cryocooler. Figure 5 schematically represents the components used for sizing a CFC in this work. Since CFC's hold fluid in a physiosorbed state, only power-based cooling was assumed. It is assumed the blanket can be charged to reach a physiosorbed density of N_2 on par with liquid N_2. Since storage capacity is inversely proportional to storage temperature, to achieve this density it was assumed the CFC is held at the boiling point of N_2 (-196°C) but at atmospheric pressure (14 psia). The cryogenic tank surrounding the blanket and aluminum is sized with the MATLAB cryogenic sizing tool, assuming liquid N_2 as the cryofluid. The cryocooler is assumed the same as listed for cryogenic N_2 options listed above and functions for the CFC as both a cooler and heater, depending on the need.

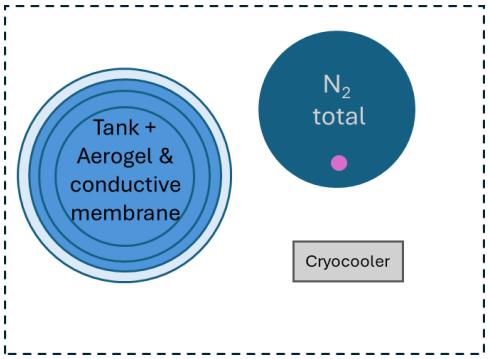


Figure 5. Schematic sizing for CFC.

5. Harvesting N_2 on Mars

Harvesting N_2 from the Martian atmosphere is a supply method of particular interest for future Mars missions because it could significantly reduce logistical mass. The Martian atmosphere, while composed primarily of CO_2 (95%), also contains 2-3% N_2, thereby presenting an ISRU opportunity if the N_2 can be captured and isolated from the CO_2 and other remaining gases (~2% Argon and trace gases, including CO). Potential gas separation techniques take advantage of the respective boiling points, molecular weights, or molecule sizes in order to distinguish inert gases from one another. Methods such as cryogenic distillation, preparative gas chromatography, and membrane separation utilize each of these properties, respectively, but none have been developed for such an application. For this work, a preliminary sizing assessment was conducted on a preparative gas chromatography system to determine the potential of harvesting N_2 with such a system during Mars surface missions when compared against other possible supply methods. Due to limited information, only the mass and volume of the system was compared. The results of this analysis are presented separately from the previous supply methods.

Preparative gas chromatography is a form of gas chromatography (GC) in which gases are isolated based on their travel time through a long capillary tube and captured accordingly. This method requires two primary units: a GC that separates the gases and a preparative collector that captures them. Though no technology development has been conducted on preparative GC for this application, it was chosen for this work because it has a significantly lower energy consumption & initial cost than cryogenic distillation and is better at separating the similarly sized inert gases found in the Martian atmosphere (i.e., N_2 and Argon) than membrane separation. Current work with preparative GC is limited to bench scale work in research laboratories. An Agilent 8890 GC [16] and GERSTEL preparative fraction collector [17] was assumed for this work. The units were sized up using the 60/40 rule [18], where 60% of the unit's

mass and volume scales linearly with flowrate and 40% remains constant. Since the required flowrate will vary based on mission needs, flowrate was parametrically varied and the resulting mass and volume recorded. No other components were considered at this time. The time to capture the required N_2 was also computed to compare processing time to unit size. The necessary gas to process for each Mars surface mission was determined through an ideal gas calculation. Average temperature and pressure conditions on Mars were assumed at 210 K and 6mBar, respectively. In order to estimate maximum yields achievable, the Martian atmosphere was assumed to contain 3% N_2 by volume.

C. Equivalent System Mass (ESM) Analysis

Barring the option of harvesting N_2 from the Martian atmosphere, all other N_2 supply options considered were compared against one another using an equivalent system mass (ESM) analysis. An ESM analysis takes disparate sizing characteristics and converts them into a single metric (mass) for easy comparison. Conversions are accomplished through the multiplication of equivalency factors (γ), which represent the estimated increase in mass of the habitat/vehicle necessary to house(γ_V)/power(γ_P)/cool(γ_C)/etc. the associated technology. The following ESM equation was used in this work:

$$\mathrm{ESM} = M_S + \gamma_V V_S + \gamma_P P_S + \gamma_C P_S - M_{O2/H2} \quad (1)$$

Here, mass (M_S), power (P_S), volume (V_S), cooling, and the mass of additional products ($M_{O2/H2}$) are accounted for in the ESM. Crew time was not included in this work due to lack of insight into maintenance and logistical coordination necessary to operate every N_2 supply system considered. For cooling, only electrical power was considered. For generation supply methods, the mass of O_2 or H_2 ($M_{O2/H2}$) produced was subtracted from the overall ESM to estimate the ECLSS benefit to providing O_2 that could be used as metabolic O_2 or providing H_2 that could be used as fuel or fed into a Sabatier to reduce CO_2. When this benefit was considered, it was assumed other systems already present in the habitat/vehicle could process and handle the O_2/H_2 produced and no additional storage containers or systems were considered.

Since equivalency factors are habitat/vehicle dependent, equivalency factors for each habitat/vehicle were obtained for the respective vehicle or nominal baseline values were used [3]. Volume equivalencies assume the N_2 supply methods will be stored outside of the vehicle/habitat. If the vehicle traverses locations, the N_2 supply method is assumed externally attached to the vehicle ($\gamma_V = 0$). If the habitat remains in a single surface location, the N_2 supply option is assumed to be housed in an unpressurized module nearby ($\gamma_V = 18.6$). The only exception to this rule is when considering missions for Orion, where the N_2 supply method is assumed housed in the European Service Module. Power and cooling equivalencies were based on the vehicle/habitat's known or expected power and cooling systems and therefore varied across vehicles/habitats. In most cases, solar photovoltaics and flow through radiators were assumed as the primary power and cooling systems, respectively. Only on the Mars surface was the primary power system assumed to be nuclear power.

III. Results

The results of this trade study are presented in the following sections. The first (A) focuses on the preliminary results of harvesting N_2 on Mars and the second (B) reviews the ESM comparison of the compressed gas tanks to identify a baseline. That baseline is then used to compare all other options to in section three (C) and four (D), where C focuses on each individual option and D brings them all together for a broader comparison. For the sections which present and discuss ESM results, each supply option's ESM is formatted as a best fit line to the total data set which considers all vehicles, habitats and associated missions, and is plotted with respect to the associated N_2 requirements. N_2 requirements provided/estimated range from around 9 kg to upwards of 4,600 kg (see Table 4). The lower N_2 requirements typically correspond to more near-term missions on transit vehicles, while the higher N_2 requirements typically correspond to surface missions on the Moon or Mars. Though the results are discussed for the entire N_2 requirement range, N_2 supply option ESMs are shown for the most relevant ranges respective to other technologies for data clarity (i.e., at low N_2 requirements and when crossover occurs).

A. Preliminary Sizing for N_2 ISRU on Mars

The lowest considered N_2 requirement for a Mars surface habitat mission was initially used to size the preparative GC system for harvesting N_2 from the Martian atmosphere. This requirement is 119 kg, corresponding to

~412,000,000 L of Martian atmosphere at the assumed conditions. Assessing for a wide range of flowrates, the system mass, volume and respective time to harvest the N_2 for this specific mission case is presented in Table 2.

Because the system is sized according to the required flowrate, it follows that the system mass and volume increases with flowrate. Conversely, time to harvest is inversely related to flowrate so an increase in flowrate decreases the time to harvest. The original expectation was that a sweet spot might be found where harvesting time and system size are relatively low. However, the mass and volume for the other considered N_2 supply options is significantly lower than those for the preparative GC system at semi-reasonable harvesting times (<5 years); referring to a scenario where the technology was sent several years ahead of the mission. The next highest mass and volume option by contrast is liquid N_2 using boil off to cool, which has a system mass of 1,409 kg and a system volume of 1.9 m^3. Both sizing parameters for liquid N_2 are therefore an order of magnitude less than the 100 L/min flowrate case for the preparative GC system that still takes almost 8 years to harvest all the N_2 needed. Furthermore, the system mass calculated for the preparative GC unit does not include additional components like a pump, compressor, or storage vessels that would add mass and volume, in addition to power which was not addressed in this analysis. For this reason, such a system was not considered viable at this time. However, this process and others (cryogenic distillation, membrane separation, etc.) may prove more promising with additional development toward space application and should not yet be discounted because of the benefit in using resources available at the mission location. Additionally, this analysis considers atmospheric extraction, but further investigation into other N_2 sources on Mars (e.g., nitrates in soil [19]) and the associated methods of extraction may provide another avenue for ISRU of N_2.

Table 2. ISRU of N_2 on Mars preparative GC sizing results.

Flowrate (L/min)	System Mass (kg)	System Volume (m^3)	Time to harvest (years)
0.01	26	0.08	78,450
0.1	43	0.14	7,845
1	207	0.66	784.5
10	1,854	5.9	78.5
100	18,324	58.2	7.8
500	91,524	291	1.6

B. Compressed Gas Tanks - Baseline

The ESM of the four different compressed gas tanks plotted as a best fit line with N_2 requirements for the missions considered, is presented in Figure 6. The color-coded shading is representative of each best fit line's standard error. Since only full tanks are considered and capacities are pre-defined, this standard error is indicative of the step-size (or tank size) of the raw data. As can be seen, the four types of tanks are not easily distinguished from one another at low N_2 requirements but as N_2 requirements increase, their differences become more apparent. The supply option based on the HPGC tank has the lowest ESM at these higher requirements. However, it is depicted with a dotted line because it is a general concept tank that is not currently planned for any future mission. Since the compressed gas option chosen will stand as the baseline for all other options, it was decided that a system based on a currently planned tank should be chosen. Therefore, ARDE was chosen as the baseline to compare all other N_2 supply options to since it has the next lowest ESM as N_2 requirements increase.

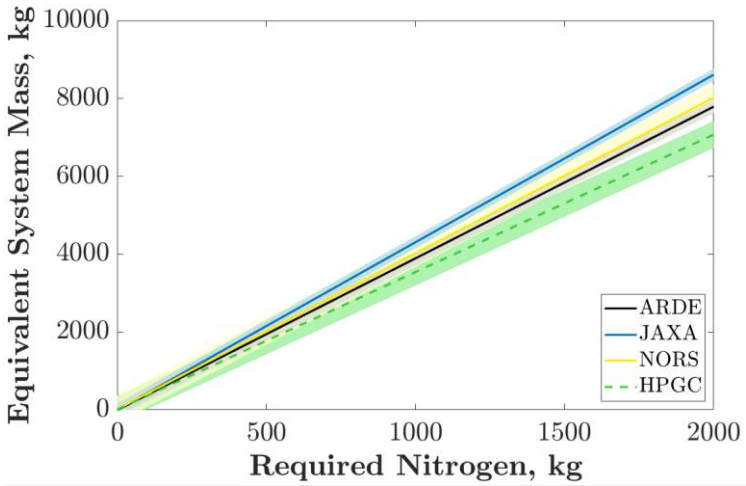


Figure 6. Compressed Gas Tank ESM Comparison.

In considering the ARDE tank system as the baseline it is important to note that the ARDE tank system sizing includes packaging based on ISS operations, while other supply options do not due to the difficulty in accurately sizing "packaging" equivalents. For example, cryogenic tanks are typically held within a vehicle using supports, rather than protected with foam, and estimating such supports requires details of the launch vehicle and location of the supply system within the vehicle. This presents a potential sizing discrepancy where the options with cryogenic tanks may be undersized as compared to the baseline compressed gas tanks. To account for this unknown potential discrepancy, the baseline was modified to include an upper and lower limit to compare all other N_2 supply options against. The upper limit refers to the ARDE system where packaging is included and the lower limit refers to the ARDE system if no packaging was considered, i.e., tanks only. A graph of how the upper and lower limit are presented in the following sections is displayed in Figure 7.

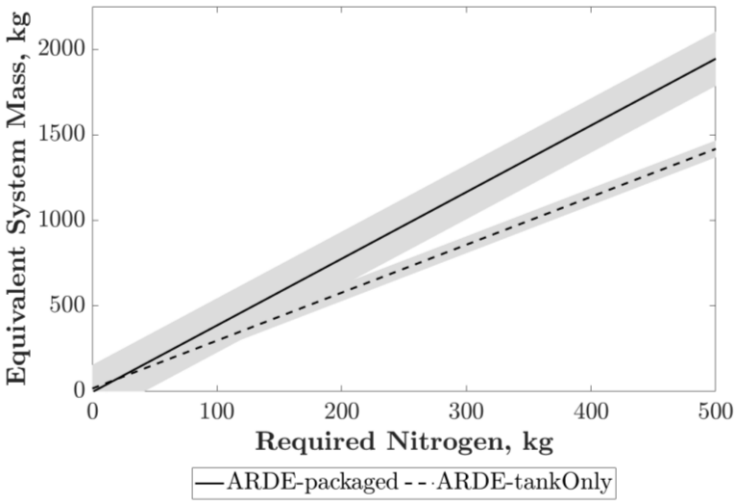


Figure 7. Upper and Lower Limit for ARDE Baseline.

C. N_2 Supply Options – Baseline Comparison

The ESM of each N_2 supply option is compared against the upper and lower limit ARDE tank option ESM with respect to required N_2 for the missions considered. The N_2 range corresponding to where the N_2 supply option of interest crosses the lower and upper limit of the baseline is referred to as the breakeven range. Table 3 in Section D summarizes the breakeven range for all options and can be referenced throughout the discussion in this section. In most cases, the ESM comparison is based primarily on system mass. Mass is the primary contributing factor to the ESM for all supply options considered, contributing on average to more than 50% of the value. Power follows closely behind in some cases, but on average contributes less than or equal to ~20% of the ESM, and volume makes up the difference. Volume is the least contributory factor because the N_2 supply options are assumed to be stored outside where volume impacts to the vehicle/habitat are significantly reduced and/or removed altogether.

1. Liquid N_2

The ESM for a liquid N_2 system is split up into the two operational concepts that were assumed to cool liquid N_2, i.e., power via a cryocooler or boil-off via loss of fluid. For the option with power, liquid N_2 begins to trade more favorably than ARDE tanks from 66 kg of N_2 (upper baseline limit) to 98 kg of N_2 (lower baseline limit), as shown by Figure 8. Conversely when liquid N_2 is cooled via boil-off, it does not trade favorably until 413 kg of N_2 up to 708 kg of N_2, as shown in Figure 9. This significantly higher breakeven range is because boil-off cooling requires not only the N_2 for the mission, but also N_2 that will be boiled off and lost to the environment. This additional N_2 mass requires the entire system be sized up, resulting in a notably larger system than if the same mission were sized with

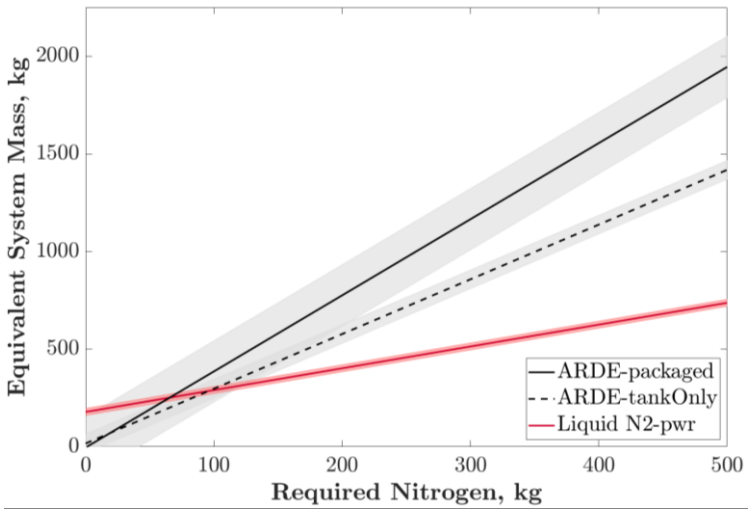


Figure 8. ESM comparison for Liquid N_2 with power for cooling.

cooling via power. The amount of boil-off also depends on the mission duration and environmental conditions. Since liquid N_2 has a very low boiling point of -196°C, there are few environments that do not exceed this temperature, and none analyzed in this work. Therefore, for the long duration missions, and particularly those for vehicles/habitats that reside in warmer environments, a significantly larger system is sized to maintain the N_2 for the mission, even if the N_2 requirements for the mission are low. This can mean that majority of the N_2 brought is lost to the environment via boil off.

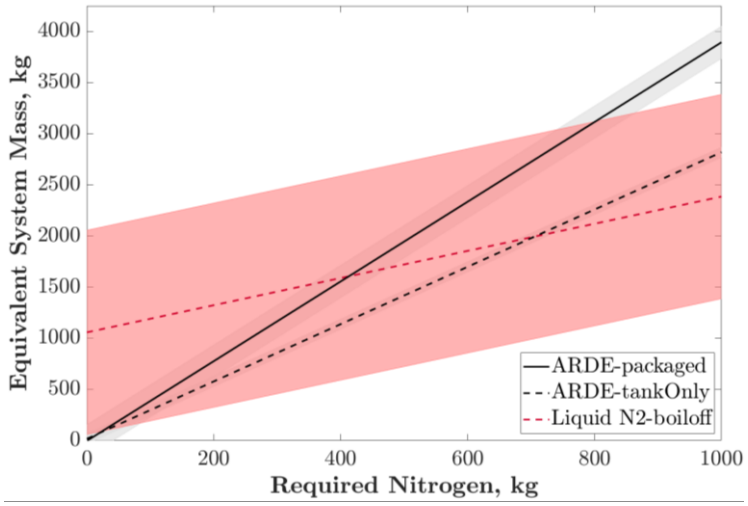


Figure 9. ESM Comparison of Liquid N_2 using boil-off for cooling.

This high sensitivity to mission duration and location also explains the significantly larger standard error (shading) for the boil off system as compared to that with power. For the same N_2 requirements, a boil-off system in a warmer environment will be appreciably larger than one in a colder environment and create a larger spread of the data. This means liquid N_2 with boil-off has the potential to trade better much earlier or much later, depending on the specific mission scenario. However, since future missions that would utilize this technology are likely to be long-duration, liquid N_2 with boil-off should not be considered if power can instead be used. When comparing the breakeven range of liquid N_2 with boil-off against all other N_2 supply options considered, liquid N_2 using boil-off to cool trades the worst. For this reason, it is not shown in further comparisons.

2. Supercritical N_2

Supercritical N_2 (SCN_2) begins to trade better than the baseline as early as 28 kg of N_2 up to 37 kg of N_2, as shown in Figure 10. Since only power can be used to cool SCN_2, the overall system ESM scales linearly with N_2 required, containing minimal standard error across all mission cases considered. In comparing SCN_2 with liquid N_2 with power, SCN_2 trades better earlier because at lower N_2 requirements, the lack of a pump and vaporizer for SCN_2 results in a net decrease in mass and power even though SCN_2 requires a higher pressure, and therefore a thicker and heavier tank. Furthermore, SCN_2 is maintained at a higher temperature (-145°C) than liquid N_2, providing additional savings on power. However, these initial benefits are overcome by the heavier tank as N_2 requirements increase, and SCN_2 begins to trade worse than liquid N_2 with power at N_2 requirements above approximately 615 kg.

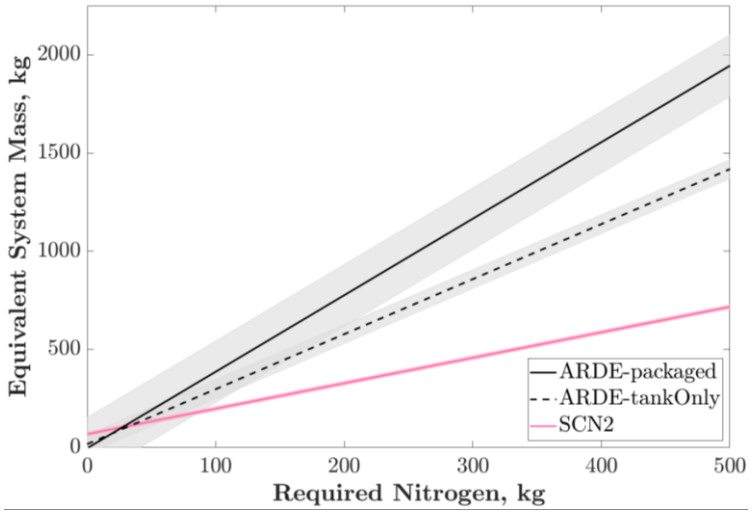


Figure 10. ESM Comparison of supercritical N_2 (SCN2).

3. N_2 Generation – Liquid N_2O

The ESM for a liquid N_2O decomposition system, the data is again split up according to the operational concept of cooling since either power via a cryocooler or boil-off via loss of fluid can be utilized. For the power option, liquid N_2O begins to trade more favorably than the compressed gas baseline starting at 62 kg of N_2 up to 92 kg of N_2, as shown by Figure 11. This range is almost identical to the breakeven range for liquid N_2 with power. Though the two systems are quite different, this similarity is due to similarities in mass, power and volume between each system's tank and additional components. Liquid N_2O is denser than liquid N_2, resulting in similarly sized tanks for the same N_2 requirement. Furthermore, the additional mass in O_2 for the liquid N_2O system is subtracted from the ESM as a benefit to the mission. The mass, power, and volume of the production unit and cryocooler for the liquid N_2O tank also happen to be similar to the mass, power and volume of the cryocooler, vaporizer and pump for the liquid N_2 tank.

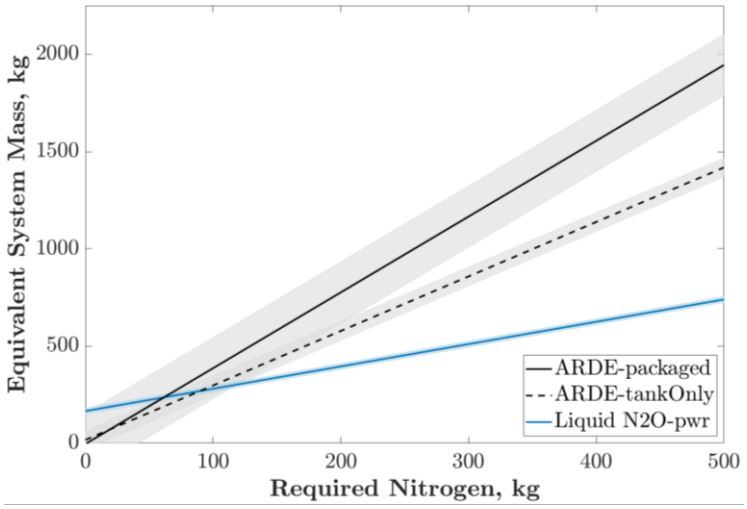


Figure 11. ESM Comparison of liquid N_2O with power for cooling.

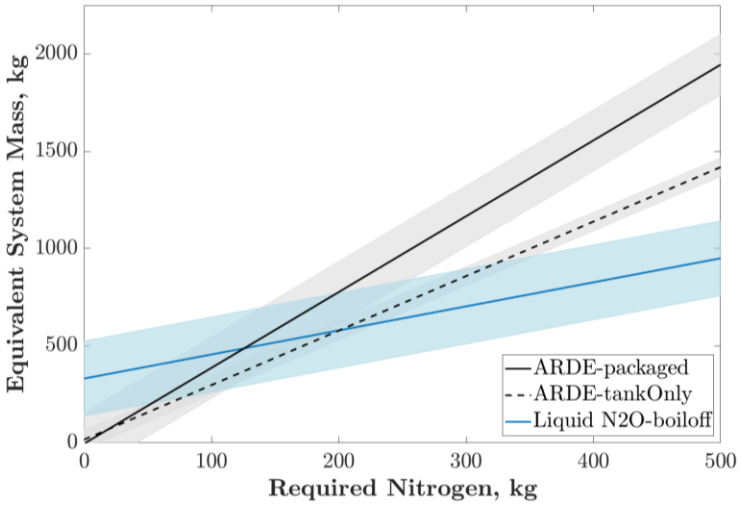


Figure 12. ESM Comparison of liquid N_2O with boil-off for cooling.

For the liquid N_2O system with boil-off, the breakeven range occurs later than the power option, occurring from 126 kg of N_2 to 203 kg of N_2, as shown in Figure 12. Much like liquid N_2 with boil-off, this same cooling method for liquid N_2O results in a larger system because additional N_2O must be provided for boil-off. Additionally, the spread of the data (shading) is also larger than the liquid N_2O system with power because of this type of cooling method's sensitivity to mission duration and environment. As a result, liquid N_2O with boil-off trades the second worst of all N_2 supply options considered, only beating out liquid N_2 with boil-off. For this reason, it is also not considered in further comparisons. However, the breakeven range and standard error for liquid N_2O with boil-off is still much reduced from liquid N_2 with boil-off. This reduction in sensitivity and system mass is a direct result of the significantly higher boiling point of N_2O (-89°C).

4. N_2 Generation – Liquid N_2H_4

The ESM for a liquid N_2H_4 decomposition system can be described by a single line with respect to N_2 requirements, as shown in Figure 13. Though this method can be cooled via power or boil-off, the boiling point of liquid N_2H_4 (114°C) is so high that power and boil-off data sets minimally differ across the mission cases considered. The liquid N_2H_4 decomposition system begins to trade better than compressed gas starting at 82 kg of N_2 up to 124 kg of N_2, making it the latest option to trade well when comparing to other systems relying on power but the earliest when comparing to other systems relying on boil-off.

Despite trading later than other systems using power to cool, liquid N_2H_4 system's breakeven range that includes boil-off cases is only ~20-30 kg of N_2 above liquid N_2 and liquid N_2O with power. This is primarily because this system in boil-off mode requires very little to boil-off to maintain a liquid state. Furthermore, in power mode, while the production unit for liquid N_2H_4 decomposition has high power requirements, the system's H_2 benefit, liquid N_2H_4's higher density as compared to liquid N_2, and the significant reduction in power to maintain a liquid state, all contribute to lowering the total ESM to within range of liquid N_2 and liquid N_2O with power. As a result, the liquid N_2H_4 system trades the best out of all options regardless of cooling mode as N_2 requirements increase. However, this result does not consider other factors, such as safety or technology maturation, which negatively impact the liquid N_2H_4 system.

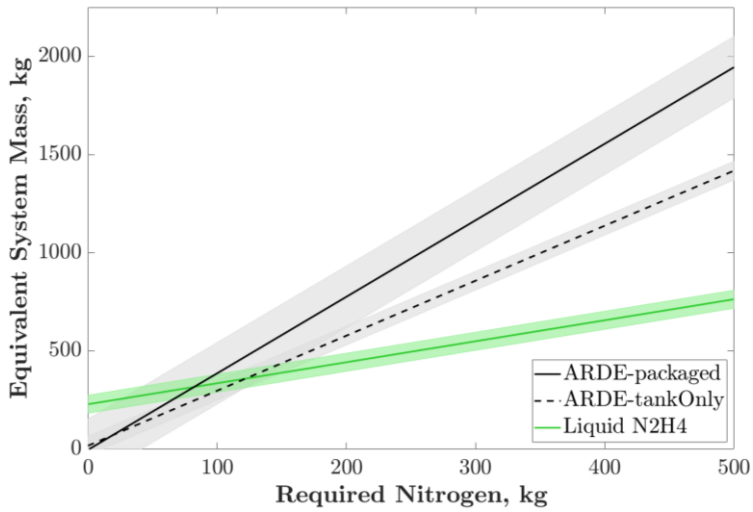


Figure 13. ESM comparison of liquid N_2H_4.

5. CFC

A CFC can be cooled via power only, so the system's ESM is described by a single line with respect to N_2 requirements, as shown in Figure 14. The CFC system's ESM begins to trade better than compressed gas tanks at 30 kg of N_2 up to 40 kg of N_2, almost identical to SCN_2. Though the CFC requires a lighter tank than SCN_2 because of the CFC's atmospheric pressure requirements, the aerogel blanket and conductive membrane make up the difference in mass. This option trades more favorably at low N_2 requirements as compared to liquid N_2 with power because a CFC does not need a vaporizer or pump, reducing the total mass, power and volume required for this option. However, much like the thicker SCN_2 tank, the aerogel blanket and thermally conductive membrane of the CFC scales linearly with N_2 requirements causing the system to trade worse than liquid N_2 with power above approximately 480 kg N_2.

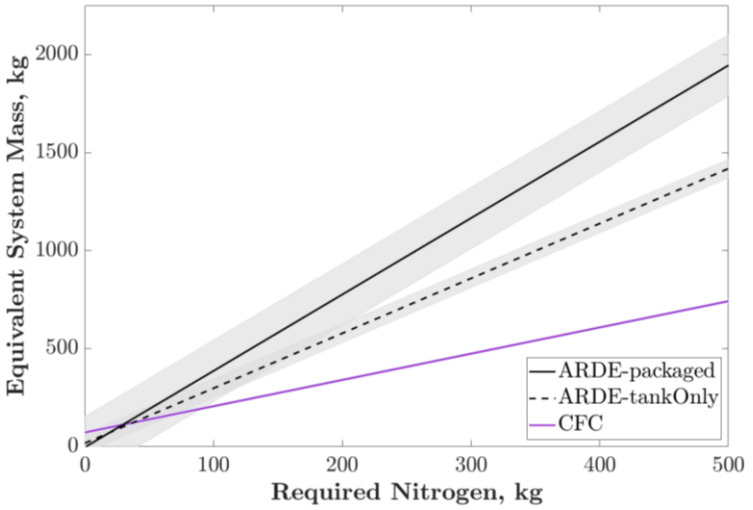


Figure 14. ESM comparison of CFC.

D. N_2 Supply Options – Collective Results & Mission Relevance

The N_2 supply option results are presented together in Figure 15, showcasing their collective comparison to the compressed gas baseline for the entire N_2 range. To provide further clarity between N_2 supply methods at lower N_2 values, the inset of Figure 15 showcases the primary point where CFC and SCN_2 begin to trade worse than the rest. The liquid N_2 and liquid N_2O systems using boil-off have been left off here since their power counterparts trade significantly better, as mentioned previously and explicitly shown here in Table 3. All options presented trade better than compressed gas tanks for N_2 requirements greater than ~124 kg of N_2 and provide increasingly better ESMs as N_2 requirements climb beyond this breakeven value. Therefore, whether a vehicle/habitat will benefit from a supply option other than compressed gas is dependent on whether N_2 requirements of relevant vehicles exceeds this breakeven value of 124 kg and by how much.

Table 3. N_2 supply options breakeven range.

N_2 Supply Method	Breakeven Range (kg)	
	Lower	Upper
SCN_2	28	37
CFC	30	40
Liquid N_2O – pwr	62	92
Liquid N_2 – pwr	66	98
Liquid N_2H_4	82	124
Liquid N_2O – boil-off	126	203
Liquid N_2 – boil-off	413	708

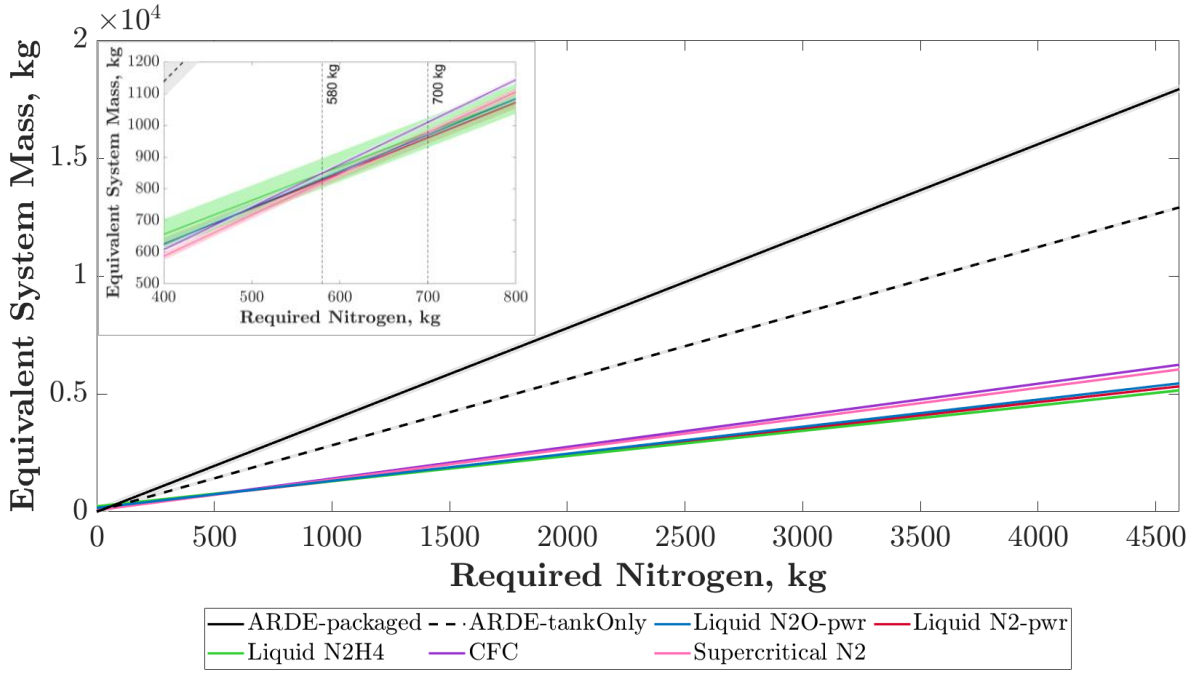


Figure 15. ESM comparison of remaining N_2 supply options up to 1000 kg of N_2 required.

Table 4 presents the considered missions and vehicles/habitats within 3 primary categories to showcase the respective N_2 usage range. Transit missions are those associated with Orion, Gateway, and the Mars transit vehicle, Lunar surface missions encapsulate HLS, PR, and MPH, and Mars surface missions refer to the Mars surface habitat. As can be seen, none of the transit missions considered exceed 300 kg of N_2 and there are missions which exist well below 124 kg, suggesting that transit missions may benefit best from continued use of compressed gas tanks. This is particularly true for short-term and/or near-term missions which primarily occupy the lower end of that range. However, for higher N_2 requirement transit missions, particularly those to Mars, another supply method may provide greater benefit.

Table 4. N_2 usage range of missions considered.

Missions	N_2 usage Range (kg)	
	Lower	Upper
Transit	9	295
Lunar surface	19	3623
Mars surface	119	4620

For surface missions, it is a different story. Not only is length of mission a factor, but EVA frequency plays a large role in N_2 usage. Surface missions on the Moon/Mars are expected to conduct a significantly greater number of EVAs

than transit missions. This greater number of EVAs on the surface leads to a greater number of airlock cycles, which leads to higher N_2 losses. Therefore, the majority of the surface missions considered sit well above the breakeven value of 124 kg, with many being an order of magnitude higher, as showcased by the upper bounds (Table 4). This means that all vehicles/habitats operating on the lunar and martian surface may benefit greatly from using an N_2 supply option other than compressed gas, especially as mission durations increase. Short and near-term missions, particularly on the lunar surface, could remain with compressed gas since many are within the lower range where compressed gas trades better, but those planning for long-term habitation should consider other options. It's also important to note briefly that whether another option should be considered for surface missions also relies heavily on the concept of operations of the airlock, i.e. whether air save is incorporated or not. While air save incurs costs related to pumping down and storing the gas (a trade that is worth conducting in future work), its use greatly influences N_2 usage. As can be seen in Table 5, the Mars surface habitat N_2 usage range differs by an order of magnitude when air save is used versus when it is not. Both vehicles/habitats on the Moon and Mars should also consider this factor for future N_2 usage and subsequent supply method.

Table 5. N_2 usage range of Mars surface missions with and without air save.

Airlock concept of operations	N_2 usage Range	
	Lower	Upper
Air save	119	729
No air save	1092	4620

The N_2 supply method which will provide the greatest benefit on future missions with N_2 requirements above 124 kg depends on exactly how high the mission requirements are and other factors not considered in this trade (e.g., crew time and maintenance). CFC and SCN_2 systems (pink and purple lines) trade the best at low N_2 (>124 kg & <580 kg) requirements but eventually cross over, trading the worst at higher N_2 requirements (>700 kg), see inset of Figure 15. Liquid N_2, liquid N_2O and liquid N_2H_4 systems are not as easily distinguished because they all trade similarly, but liquid N_2H_4 systems do have a slight edge in the trade at the higher N_2 requirements considered (>1000 kg). However, there are safety concerns associated with storing liquid N_2H_4 and its technology readiness level (TRL) for conversion to pure nitrogen is quite low [14]. Additionally, the CFC system is also a low TRL technology [8], meaning future developments may improve its overall ESM but it may also take significantly longer to develop as compared to higher TRL technologies such as liquid N_2 and liquid N_2O. Therefore, since long duration missions with high EVA frequency are likely to benefit most from a non-compressed gas option, this study culminated in the recommendation that liquid N_2 with power and liquid N_2O with power should be pursued in developmental efforts as they trade best at higher N_2 requirements, have higher TRLs, and fewer safety concerns of the options considered.

IV. Conclusion

This N_2 supply trade study identified, sized and traded potential N_2 supply methods across the evaluated N_2 usage requirements for all future missions for the currently planned vehicles/habitats. Based on the results of this trade, it is clear that there is no "one size fits all". At small N_2 requirements and near-term missions, compressed gas is likely still the best option as it trades the best and requires no further development to employ. This primarily refers to transit vehicles for this work. However, as N_2 requirements increase above the breakeven range of ~124 kg of N_2, missions are likely to benefit from using a method utilizing higher density fluids instead of compressed gas tanks. This benefit becomes significant on long duration surface missions where high EVA frequencies result in large N_2 needs, often an order of magnitude above the general breakeven range. The exact method that trades best above the breakeven range depends on N_2 requirements, as well as other factors only qualitatively considered in this work. However, based on the trade results and qualitative assessment of TRL and safety, liquid N_2 and liquid N_2O systems utilizing power as their cooling method are identified as best overall. Such supply options require forward work if they are to be realized in future spacecraft.

Future analyses in this space would be bolstered by updated parameters for missions/vehicles/habitats as they become available/are updated, thereby reducing the variable space and narrowing the expected N_2 usage range. Further development and experimental testing on the various systems considered for N_2 supply could also provide improved sizing information. For example, the power method to cool may require additional components to integrate a cryocooler with a composite tank but these components are yet to be identified. Sensitivity analyses could also aid in this effort to understand the most impactful sizing factors. Lastly, the inclusion of spares, maintenance parts, and an estimate of crew time would also increase the fidelity of the trade. Experimentation on component duration would also aid in this effort.

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